Weekly Status Report

WEEK 6

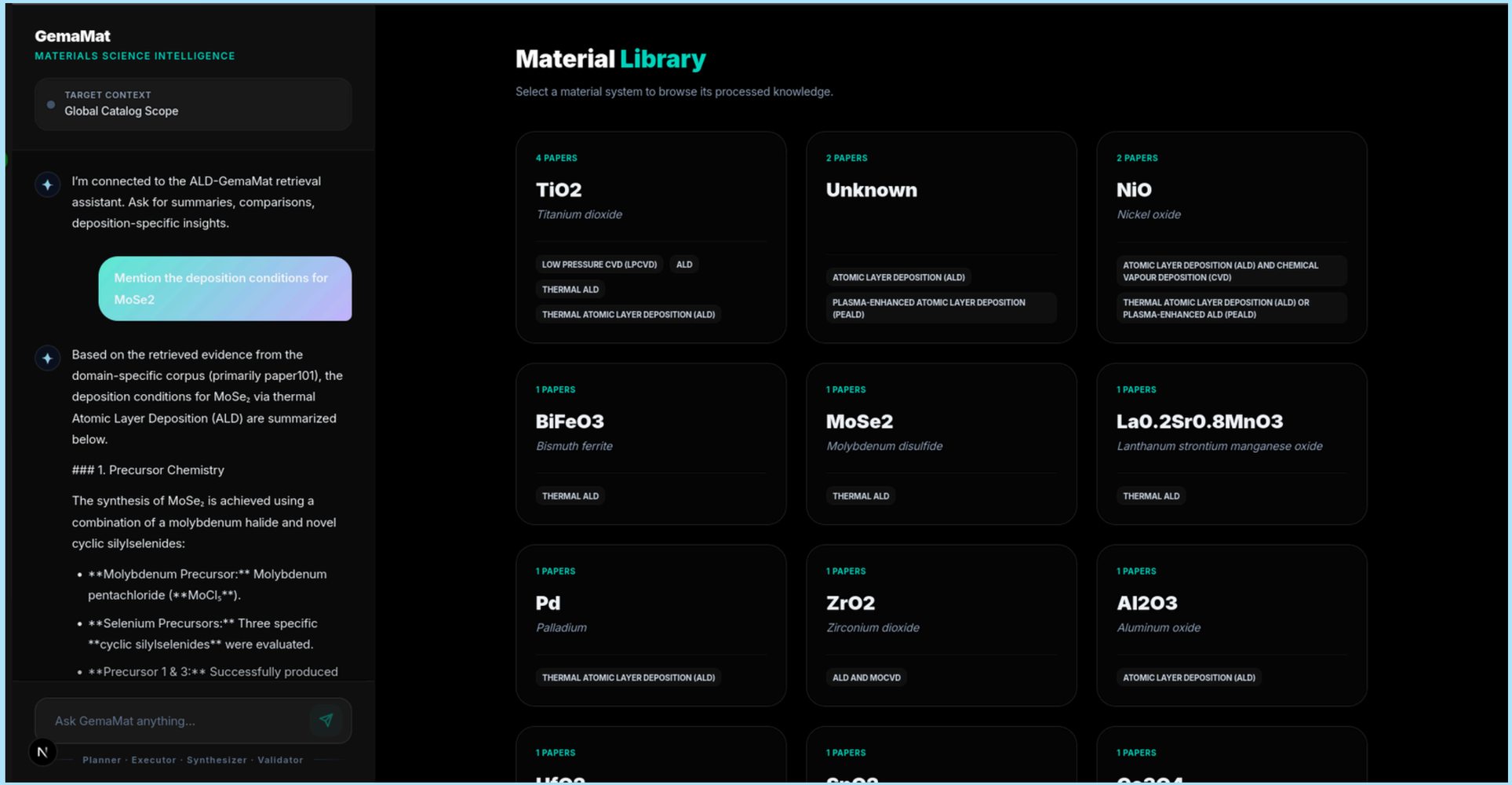
UGP1 | MSE 496

1

UPDATES TO THE FRONTEND

GENERAL STATUS

RE IMAGINED FRONTEND - COLOR CHANGE



FILTERING BY MATERIALS INSTEAD OF PAPERS

Material Library

Select a material system to browse its processed knowledge.

4 PAPERS

TiO2

Titanium dioxide

LOW PRESSURE CVD (LPCVD) | ALD | THERMAL ALD | THERMAL ATOMIC LAYER DEPOSITION (ALD)

2 PAPERS

Unknown

ATOMIC LAYER DEPOSITION (ALD) | PLASMA-ENHANCED ATOMIC LAYER DEPOSITION (PEALD)

2 PAPERS

NiO

Nickel oxide

ATOMIC LAYER DEPOSITION (ALD) AND CHEMICAL VAPOUR DEPOSITION (CVD) | THERMAL ATOMIC LAYER DEPOSITION (ALD) OR PLASMA-ENHANCED ALD (PEALD)

1 PAPERS

BiFeO3

Bismuth ferrite

THERMAL ALD

1 PAPERS

MoSe2

Molybdenum disulfide

THERMAL ALD

1 PAPERS

La0.2Sr0.8MnO3

Lanthanum strontium manganese oxide

THERMAL ALD

THREE TIER ACCESS TO INFORMATICS

Material Library

Select a material system to browse its processed knowledge.

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THERMAL ALD

← Back to Materials

TiO2 Archive

Found 4 contributions

PAPER1 — 10 METHODS
TiO2 via Low Pressure CVD (LPCVD)
Low Temperature CVD of Crystalline Titanium Dioxide Films Using Tetranitratotitanium(IV)
LOW PRESSURE CVD (LPCVD) | 184°C

PAPER1004 — 12 METHODS
TiO2 via ALD
TiO2 films were deposited onto stainless steel substrates using ALD, showing improved corrosion resistance.
ALD | 150°C

PAPER1005 — 10 METHODS
TiO2 via Thermal ALD
Conformal TiO2 layer deposited onto CrN coated stainless steel using ALD improves corrosion resistance.
THERMAL ALD | 200°C

PAPER1007 — 6 METHODS
TiO2 via Thermal Atomic Layer Deposition (ALD)
The study investigates the structural characterization of TiO2–Cr2O3 nanolaminates grown by atomic layer deposition.
THERMAL ATOMIC LAYER DEPOSITION (ALD) | 375°C

← Back to TiO2 | View Original PDF

TARGET MATERIAL	PROCESS TYPE	TEMPERATURE RANGE	CHARACTERIZATION METHODS
Titanium dioxide	Low Pressure CVD (LPCVD)	184-330 °C	10 methods

Paper Summary

Low Temperature CVD of Crystalline Titanium Dioxide Films Using Tetranitratotitanium(IV)

Tetranitratotitanium(IV) (TNT)

› Evidence

Target Material

TiO2 | Titanium dioxide | Oxide

› Evidence

2

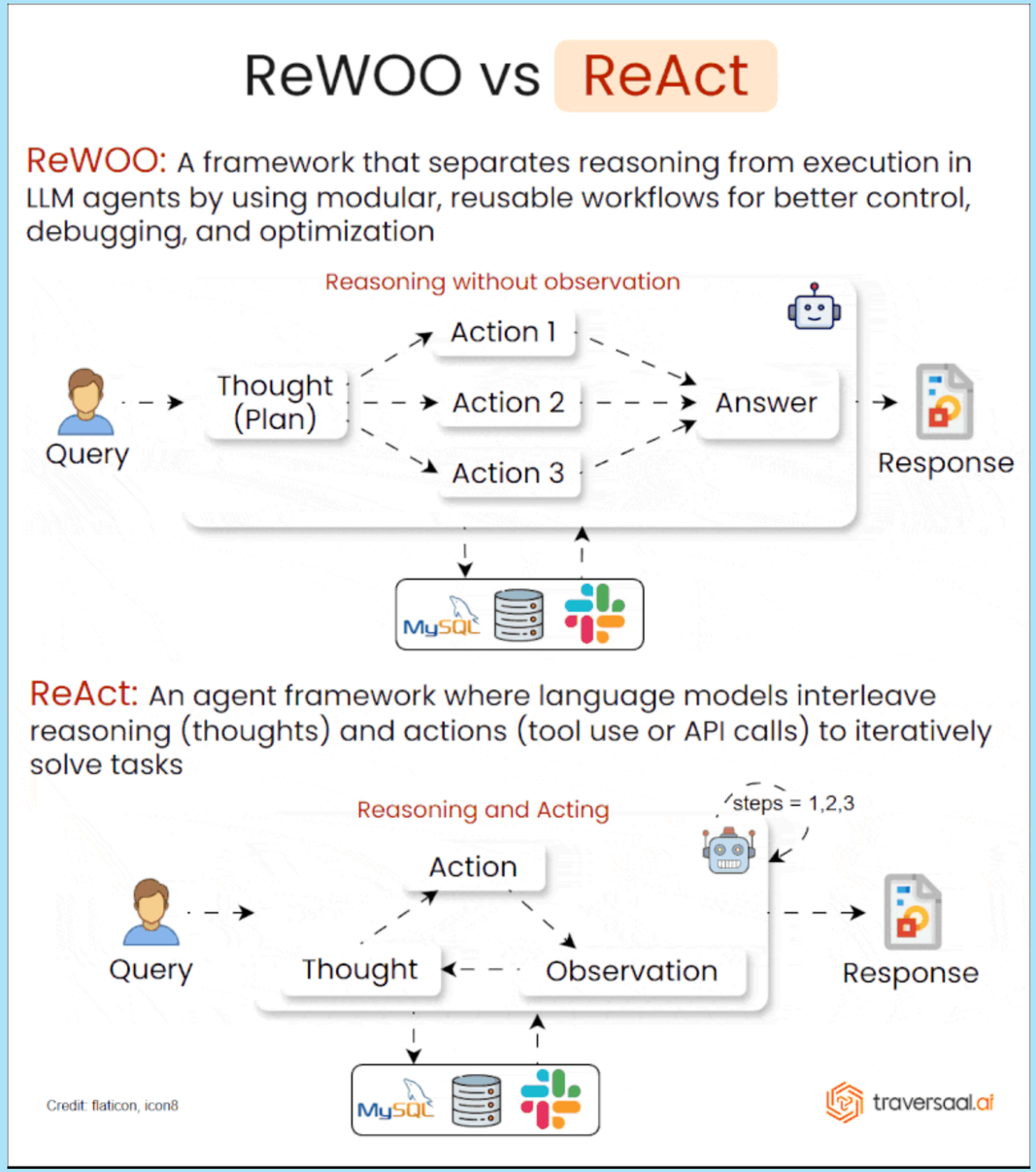
PIPELINE UPDATES

GENERAL STATUS

WHAT WE HAD EARLIER

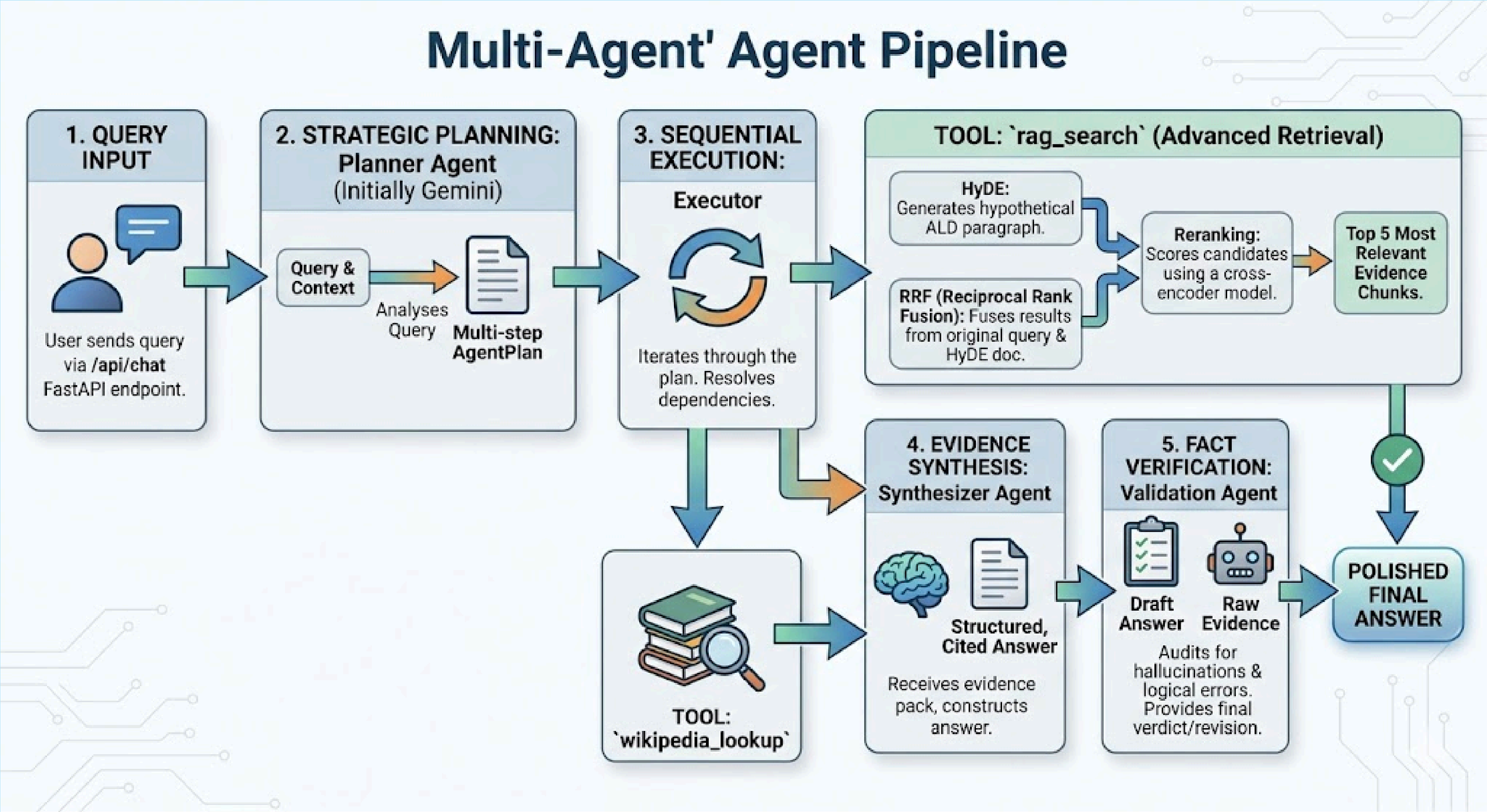
ADVANCED RAG PIPELINE USING HYDE, CHUNK FUSION AND RERANKING THE CHUNKS

PARADIGMS IN AGENTIC AI



WE EMPLOY A REWOO TYPE ARCHITECTURE HERE TO SAVE ON LLM TOKENS.

AGENTIC PIPELINE



PLANNER AGENT


```
{
  "planner_summary": "Retrieve and compare GPC data for TiO2 from TMA and TDMAT
precursors.",
  "steps": [
    {
      "step_id": "A1",
      "step_type": "analysis",
      "title": "Query Decomposition",
      "objective": "Identify comparing entities: Precursor 1 (TMA), Precursor 2
(TDMAT), Target (TiO2), Metric (GPC).",
      "depends_on": [],
      "expected_output": "Decomposition checklist."
    },
    {
      "step_id": "E1",
      "step_type": "tool",
      "title": "Fetch TMA Evidence",
      "objective": "Search ALD corpus for TiO2 deposition using TMA.",
      "tool_name": "rag_search",
      "arguments": { "query": "TiO2 GPC using TMA precursor" },
      "depends_on": ["A1"],
      "expected_output": "Grounded GPC values for TMA."
    },
    {
      "step_id": "E2",
      "step_type": "tool",
      "title": "Fetch TDMAT Evidence",
      "objective": "Search ALD corpus for TiO2 deposition using TDMAT.",
      "tool_name": "rag_search",
      "arguments": { "query": "TiO2 GPC using TDMAT precursor" },
      "depends_on": ["A1"],
      "expected_output": "Grounded GPC values for TDMAT."
    },
    {
      "step_id": "E3",
      "step_type": "tool",
      "title": "Wikipedia Background",
      "objective": "Fetch chemical properties of TDMAT to explain potential
differences in ALD window.",
      "tool_name": "wikipedia_lookup",
      "arguments": { "query": "Tetrakis(dimethylamino)titanium" },
      "depends_on": ["E2"],
      "expected_output": "Chemical background for TDMAT."
    }
```


THIS IS A SAMPLE OUTPUT OF THE PLANNER AGENT

THE EXECUTOR (NO LLM USED)

The Executor processes this JSON sequentially, but treats it as a graph using a central Result Store.

Step A1: Reasoning

- Action: No tool is called. The Executor simply records the "objective" into the Result Store under key A1.
- Result Store: { "A1": { "analysis": "Identify comparing entities..." } }

Step E1: Independent Tool Call

- Action: Executor sees depends_on: ["A1"]. Since A1 is in the Store, it proceeds.
- Tool: runs rag_search tool
- Result Store: { "A1": ..., "E1": { "sources": [...], "summary": "Found GPC ~0.5 A/cycle" } }

Step E2: Another tool call

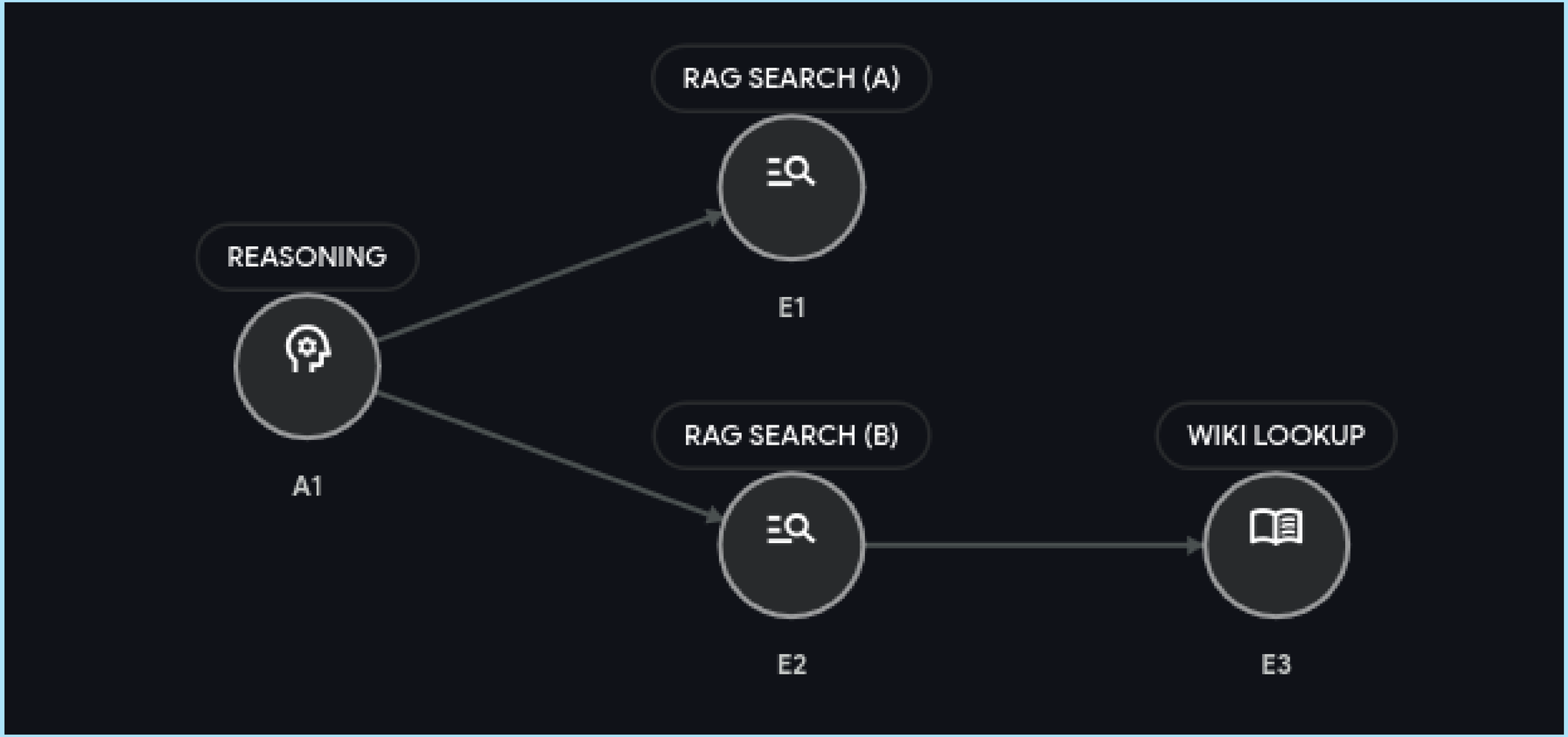
- Action: Executor runs this exactly like E1. It doesn't need data from E1, but it's part of the same "tier" of the graph.
- Result Store: { "A1": ..., "E1": ..., "E2": { "summary": "Found GPC ~0.4 A/cycle" } }

Step E3: Dependent Piped Call

- Action: Executor sees depends_on: ["E2"]
- Runs the wikipedia_lookup tool and then updates the Result Store

At the very end, the Synthesizer Agent doesn't just see the user query. It receives the entire Result Store. It sees: A1, E1, E2, E3.

DAG (DIRECTED ACYLIC GRAPH)



SYNTHESIZER AGENT

```python
def generate_answer(
    self,
    query: str,
    conversation: list[ChatTurn],
    scope_paper_id: str | None,
    plan_summary: str,
    execution_summary: str,
    context_blocks: str,
) -> str:
    scope_line = scope_paper_id or "global corpus"
    return self._generate(
        system_instruction=(
            "You are the Synthesizer Agent in an ALD (Atomic Layer Deposition) retrieval pipeline. "
            "You receive raw evidence blocks retrieved from two sources: a RAG search over ALD-specific "
            "documents, and a Wikipedia lookup. Your job is to synthesize these into a single, coherent, "
            "scientifically accurate answer to the user's query.\n\n"

            "## SYNTHESIS PRINCIPLES\n\n"

            "### 1. Source Hierarchy\n"
            "- RAG evidence is primary — always prefer RAG chunks for process-specific claims "
            "(GPC, ALD window, dose times, precursor chemistry, film properties)\n"
            "- Wikipedia is supplementary — use only for general chemistry context, definitions, "
            "or background where RAG evidence is silent\n"
            "- Never let Wikipedia override a RAG-sourced numerical or process-specific claim\n"
            "- If RAG and Wikipedia contradict, state both positions explicitly — do not silently resolve\n\n"

            "### 2. Strict Evidence Boundaries\n"
            "- Every claim you make must be traceable to an evidence block\n"
            "- Do not interpolate numerics: if evidence says >150°C, do not write 150–300°C\n"
            "- Do not infer causality beyond what the evidence states\n"
            "- Do not bridge two separate evidence facts with a causal statement unless the evidence itself makes tha
            "- If evidence is insufficient to answer part of the query, explicitly say so — do not fill gaps\n\n"

            "### 3. ALD-Specific Scientific Accuracy\n"
            "- Temperature vs GPC: in ALD, GPC typically decreases at higher temperatures due to "
            "ligand desorption or increased reaction reversibility — do not imply a CVD-like positive correlation\n"
            "- Always distinguish thermal ALD from PE-ALD (plasma-enhanced) if both appear in evidence\n"
            "- Saturation behavior: dose and purge times must be described in the context of self-limiting reactions\
            "- Nucleation: distinguish substrate-inhibited vs substrate-enhanced nucleation if evidence supports it\n
            "- Phase and stoichiometry: distinguish amorphous vs crystalline, and exact phases "
```

DabeetDas (4 days ago) Ln 164, Col 96 (2 selected) Spaces: 4 UT

VALIDATION AGENT

```python
def validate_answer(
    self,
    query: str,
    draft_answer: str,
    validation_focus: list[str],
    context_blocks: str,
) -> dict[str, Any]:
    return self._generate_json(
        system_instruction=(
            "You are the final Validation Agent in an ALD (Atomic Layer Deposition) retrieval pipeline. "
            "You receive a synthesized answer and the raw evidence blocks it was built from. "
            "Your sole job is to audit whether the synthesized answer is faithful to the evidence.\n\n"

            "## WHAT TO CHECK\n\n"

            "### Factual Grounding\n"
            "Verify every claim traces to an evidence block. Flag:\n"
            "- Numeric values absent from evidence: GPC, ALD window bounds, dose/purge times, "
            "film thickness, dielectric constant, refractive index, impurity levels\n"
            "- Precursor chemistry errors: wrong ligands, incorrect oxidant pairings\n"
            "- Material property misattributions: wrong bandgap, phase, density\n"
            "- Bulk material properties (often from general sources) substituted for thin-film ALD values\n"
            "- Temperature-GPC errors: in ALD, GPC typically decreases at higher temperatures — "
            "flag if the answer implies otherwise without evidence\n\n"

            "### Logical Consistency\n"
            "Check internal coherence across the entire answer:\n"
            "- Self-contradictions between sections\n"
            "- Saturation or self-limiting claims inconsistent with cited dose/purge behavior\n"
            "- Growth mode descriptions that contradict the stated substrate or precursor\n"
            "- Causal bridges between two evidence facts that the evidence itself never makes\n"
            "- Comparative statements (faster, thinner, more conformal) that flip direction mid-answer\n\n"

            "### Cross-Verification\n"
            "Reconcile claims across all supplied evidence blocks:\n"
            "- False consensus: answer presents agreement where evidence blocks actually disagree\n"
            "- Cherry-picking: one block used while a contradicting block is silently ignored\n"
            "- Numeric interpolation: values not in any block but derived from them (e.g., evidence "
            "says >150°C but answer states 150–300°C)\n"
            "- Characterization conclusions (XPS, TEM, XRR, ellipsometry, TOF-SIMS) that exceed "
```

Thank you